

HDOC Day-18

Menu-Driven Digit-Based Number Operations Using Switch-Case and User-Defined Functions

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Aim

To write a menu-driven C program using switch-case and four separate void user-defined functions to check whether a positive integer is a Strong, Armstrong, Harshad, or Duck number.

Algorithm

1. Start and display a menu with four choices.
2. Read the user's choice.
3. Use switch-case to call the corresponding function.
4. strong(): read a positive integer, find the factorial of every digit, add the factorials, and compare the sum with the original number.
5. armstrong(): read a positive integer, count its digits, raise each digit to that count, add the values, and compare the sum with the original number.
6. harshad(): read a positive integer, find the sum of its digits, and check whether the original number is exactly divisible by that sum.
7. duck(): read a positive integer and check whether it contains at least one zero digit.
8. Display the result. If the menu choice is invalid, display an error message.
9. Stop.

Test Case Table

No.	Option / Input	Type	Expected Result
1	Strong: 145	Valid	145 is a Strong number
2	Strong: 123	Valid	123 is not a Strong number
3	Armstrong: 153	Valid	153 is an Armstrong number
4	Armstrong: 125	Valid	125 is not an Armstrong number
5	Harshad: 18	Valid	18 is a Harshad number
6	Harshad: 19	Valid	19 is not a Harshad number
7	Duck: 1023	Valid	1023 is a Duck number
8	Duck: 1234	Valid	1234 is not a Duck number
9	Choice: 5	Invalid	Invalid choice message
10	Input: 1	Boundary	Program processes the positive one-digit input

C Program

```
#include <stdio.h>

void strong();
void armstrong();
void harshad();
void duck();

int main() {
    int choice;

    printf("----- MENU -----\\n");
    printf("1. Check Strong Number\\n");
    printf("2. Check Armstrong Number\\n");
    printf("3. Check Harshad Number\\n");
    printf("4. Check Duck Number\\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: strong(); break;
        case 2: armstrong(); break;
        case 3: harshad(); break;
        case 4: duck(); break;
        default: printf("Invalid choice\\n");
    }

    return 0;
}

void strong() {
    int n, temp, digit, sum = 0, fact, i;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input\\n");
        return;
    }

    temp = n;
    while (temp > 0) {
        digit = temp % 10;
        fact = 1;

        for (i = 1; i <= digit; i++)
            fact = fact * i;

        sum = sum + fact;
        temp = temp / 10;
    }

    if (sum == n)
        printf("%d is a Strong number\\n", n);
    else
        printf("%d is not a Strong number\\n", n);
}

void armstrong() {
    int n, temp, digit, digits = 0;
    int sum = 0, power, i;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input\\n");
        return;
    }

    temp = n;
    while (temp > 0) {
        digits++;
        temp = temp / 10;
    }

    temp = n;
    while (temp > 0) {
        digit = temp % 10;
        power = 1;

        for (i = 1; i <= digits; i++)
```

```

        power = power * digit;

        sum = sum + power;
        temp = temp / 10;
    }

    if (sum == n)
        printf("%d is an Armstrong number\n", n);
    else
        printf("%d is not an Armstrong number\n", n);
}

void harshad() {
    int n, temp, digit, sum = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input\n");
        return;
    }

    temp = n;
    while (temp > 0) {
        digit = temp % 10;
        sum = sum + digit;
        temp = temp / 10;
    }

    if (n % sum == 0)
        printf("%d is a Harshad number\n", n);
    else
        printf("%d is not a Harshad number\n", n);
}

void duck() {
    int n, temp, digit, found = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input\n");
        return;
    }

    temp = n;
    while (temp > 0) {
        digit = temp % 10;

        if (digit == 0) {
            found = 1;
            break;
        }

        temp = temp / 10;
    }

    if (found)
        printf("%d is a Duck number\n", n);
    else
        printf("%d is not a Duck number\n", n);
}

```

Sample Executions

TEST 1 - STRONG NUMBER
Enter your choice: 1
Enter a positive integer: 145
145 is a Strong number

TEST 2 - ARMSTRONG NUMBER
Enter your choice: 2
Enter a positive integer: 153
153 is an Armstrong number

TEST 3 - HARSHAD NUMBER
Enter your choice: 3
Enter a positive integer: 18
18 is a Harshad number

TEST 4 - DUCK NUMBER
Enter your choice: 4
Enter a positive integer: 1023
1023 is a Duck number

TEST 5 - INVALID MENU CHOICE
Enter your choice: 5
Invalid choice

Result

The menu-driven C program was successfully prepared using switch-case and four separate void user-defined functions. The program checks Strong, Armstrong, Harshad and Duck numbers and was tested with valid, invalid and boundary test cases.